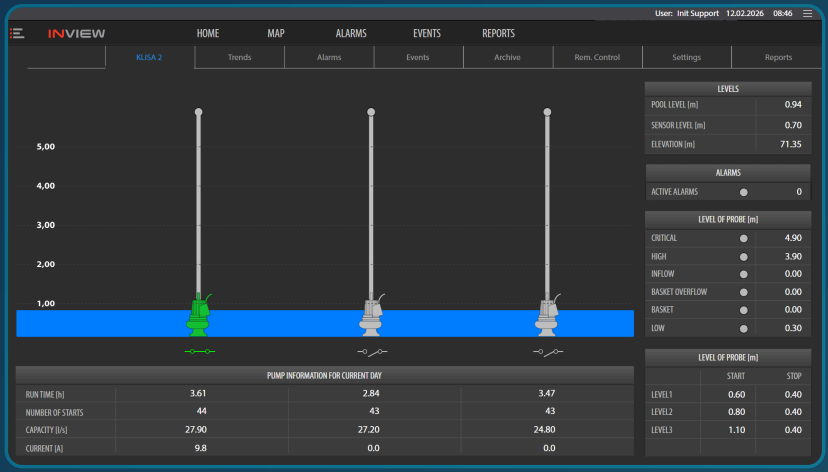
SCADA System for Water Management

Distributed Location Monitoring

Presentation Overview



What is SCADA?

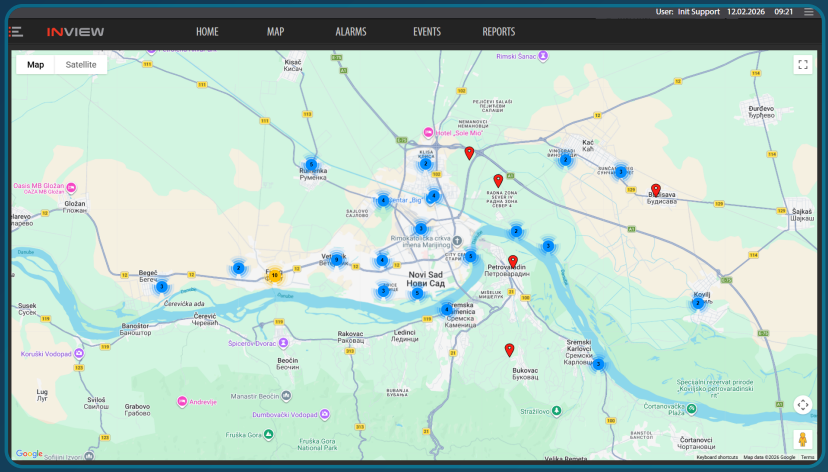
Supervisory Control and Data Acquisitionsdf sdf

- Real-time monitoring and control system
- Remote facility management
- Data collection and analysis
- Automated decision making

Key Benefits:

- 24/7 monitoring capability
- Reduced operational costs
- Improved response times
- Enhanced system reliability

Water Management Challenges



Distributed Infrastructure

- Multiple remote locations
- Vast geographical coverage
- GPR as communication

Critical Needs:

- Real-time water level monitoring
- Pressure and flow control
- Leak detection systems
- Pump station management
- Reservoir level tracking

System Architecture

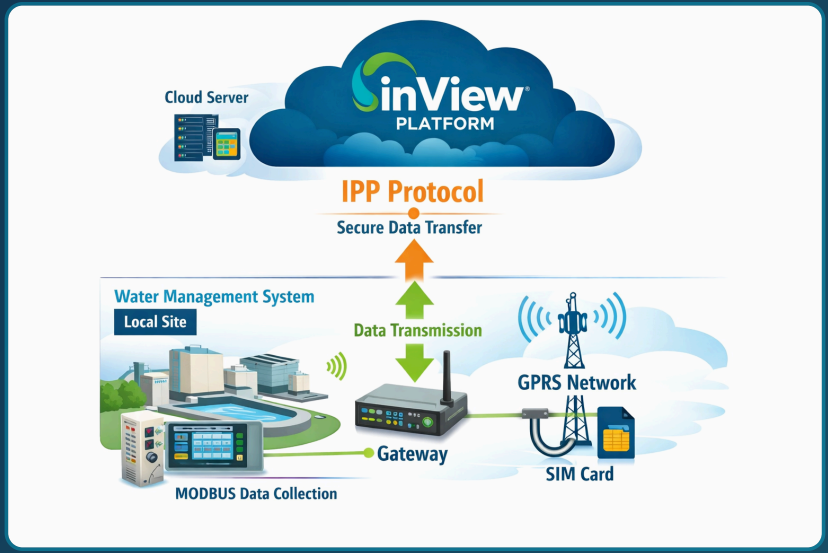
SCADA Components

Field Devices (RTUs/PLCs)

- Sensors and meters
- Actuators and valves
- Local controllers

Communication Network

- Cellular links (GPRS communication / SIM cards)
- InView gateway i51 (using IPP -> TCP sockets)
- InView gateway i3x (using I3X -> MQTT)



Web Client

Table review

- Sound alarming
- Locations overview

Single stations review

- Overview
- Trends analises
- Alarms/Events view

Editor

How is it made?

Single page components

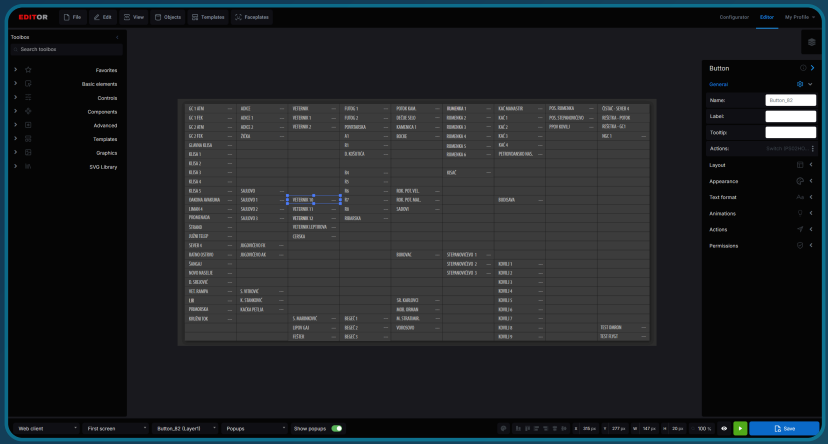
- Buttons management
- Link variables to value
- Link variables to states

Tempaltes and sufixes

- Prefix - sufix linking
- TagTree

Mobile screens

- faceplates and templates



Configurator

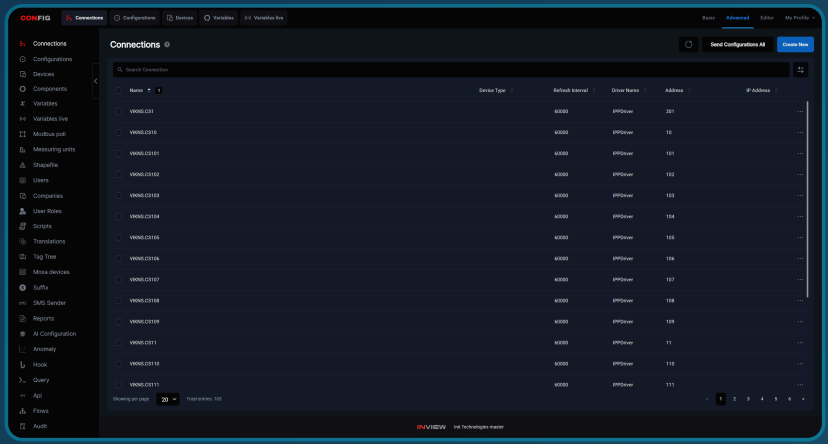
Basics

Connection management

- IPP driver
- i3x driver
- Other drivers (MQTT, OPCUA, S7)

Variable managegement

- Internal vs live
- Loggability
- Alarms and Events



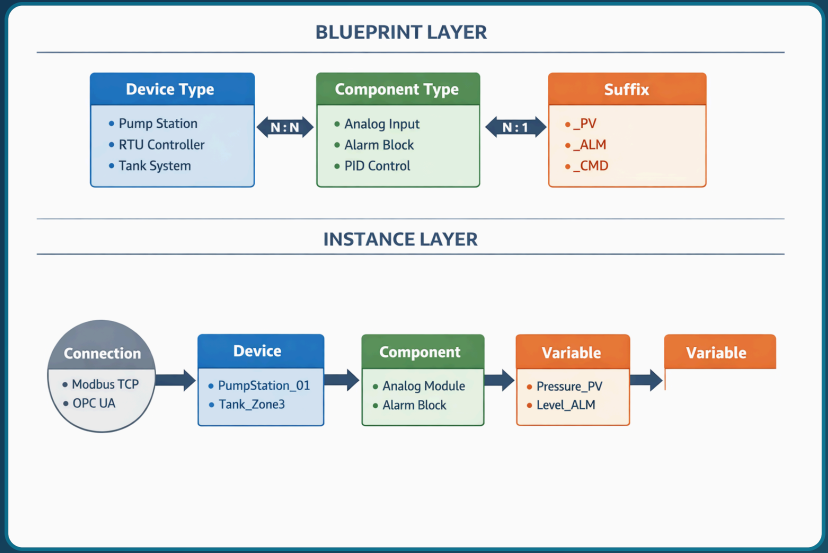
Configurator

Devices and components management

Device
- Add Device vs Connection

Component
- N-N with Device

Sufix
- Batch handle
- Unlink variable



Configurator

Scripting

Scripts

- InView C#

Edge Scripts

- i3x only

Script Templates

Questions & Discussion

Thank you for your attention! Contact Information:

- Technical Support
- Implementation Team
- Follow-up Sessions